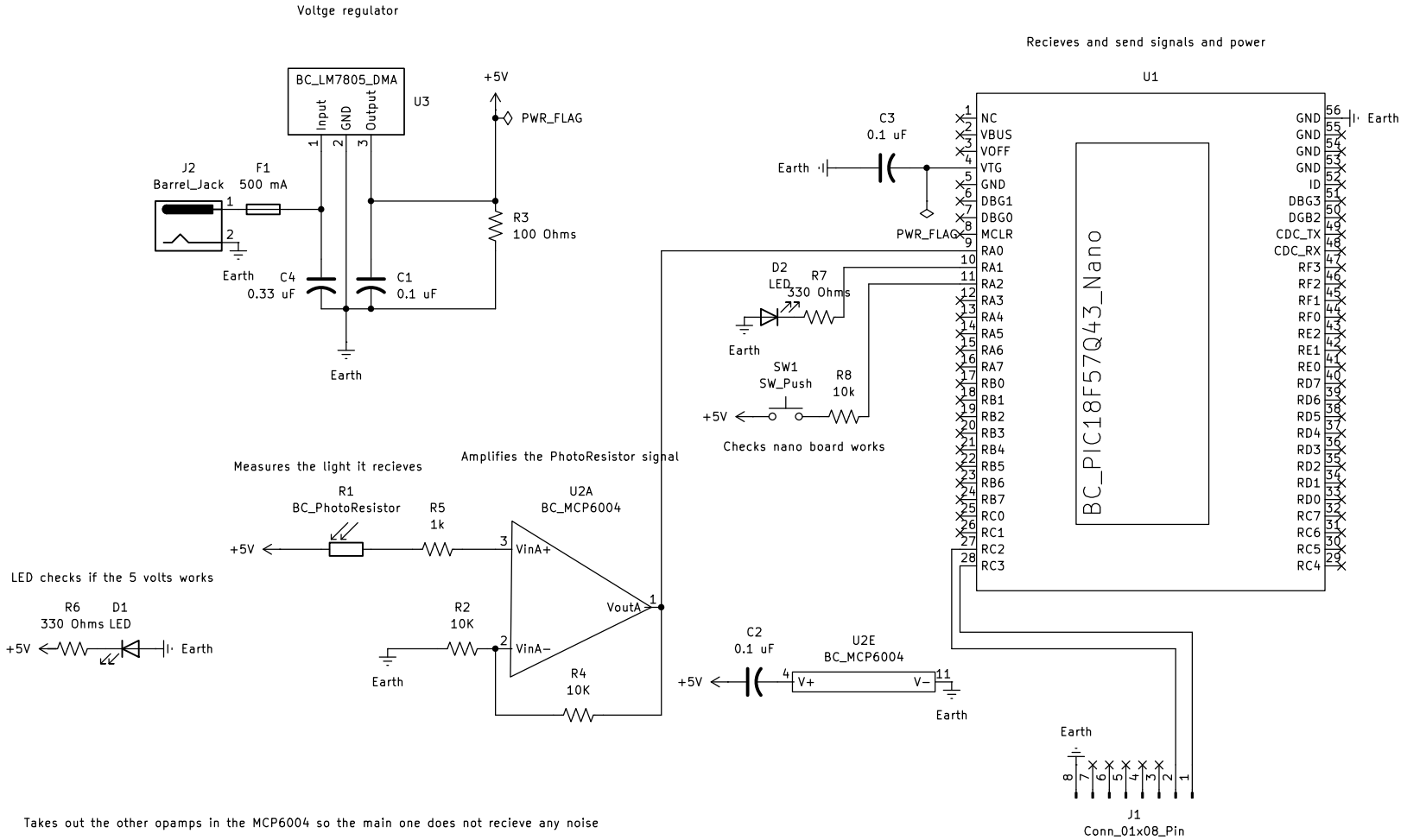
Voltge regulator
BC_LM7805_DMA
Input
GND
Output
U3
+5V
PWR_FLAG
J2
Barrel_Jack
F1
500 mA
R3
100 Ohms
Earth
C4
0.33 uF
C1
0.1 uF
Earth
Recieves and send signals and power
U1
C3
0.1 uF
Earth
PWR_FLAG
NC
VBUS
VOFF
VTG
GND
DBG1
DBG0
MCLR
RA0
RA1
RA2
RA3
RA4
RA5
RA6
RA7
RB0
RB1
RB2
RB3
RB4
RB5
RB6
RB7
RC0
RC1
RC2
RC3
BC_PIC18F57Q43_Nano
GND
GND
GND
GND
ID
DBG3
DBG2
CDC_TX
CDC_RX
RF3
RF2
RF1
RF0
RE2
RE1
RE0
RD7
RD6
RD5
RD4
RD3
RD2
RD1
RD0
RC7
RC6
RC5
RC4
Earth
D2
LED
R7
330 Ohms
Earth
SW1
SW_Push
R8
10k
+5V
Checks nano board works
Measures the light it recieves
Amplifies the PhotoResistor signal
R1
BC_PhotoResistor
R5
1k
U2A
BC_MCP6004
+5V
VinA+
VoutA
VinA−
R2
10K
Earth
R4
10K
LED checks if the 5 volts works
R6
330 Ohms
D1
LED
+5V
Earth
C2
0.1 uF
U2E
BC_MCP6004
+5V
V+
V−
Earth
Earth
J1
Conn_01x08_Pin
Connects my circuit connects to the main circuit
Takes out the other opamps in the MCP6004 so the main one does not recieve any noise
U2B
BC_MCP6004
Earth
VinB+
VoutB
VinB−
U2C
BC_MCP6004
Earth
VinC+
VoutC
VinC−
U2D
BC_MCP6004
Earth
VinD+
VoutD
VinD−